CONFIDENTIAL

High-Power Wireless Power Transfer (WPT) Coils:

— Addressing thermal physics limits under 11kW~22kW+ high-frequency operation; achieving radial heat dissipation breakthrough for coil body.

01 Industry Pain Points: Thermal Bottlenecks at High Frequency & Power

ACR Surge

At 85kHz–150kHz, skin and proximity effects reduce effective conductive cross-section; Joule heating rises nonlinearly with power.

Limited Traditional Heat Path

Litz wire + thermal paste + cold plate structures require heat to pass through multiple insulating layers, causing severe thermal resistance and hotspots.

Physical Limits of Existing Liquid Cooling

Why traditional thermal epoxy (1–3 W/m·K) no longer meets requirements

P1

Insulation Barrier

Low-thermal-conductivity polymer enamel blocks heat from escaping the copper core.

P2

Radial Obstruction

Micro-gaps and enamel contacts in stranded wire reduce effective radial thermal conductivity.

P3

Interface Contact Failure

Lack of atomic-level bonding between conductor and potting creates contact thermal resistance.

02 CuFeng Solution: Diamond-Copper Composite Conductor

Physical Evolution Logic

1. **Atomic-Level Interface Fusion**
   Proprietary thermal process achieves metallurgical bonding between micron reinforcement and copper matrix, eliminating interface thermal resistance.

2. **Multi-Level Thermal Network**
   Parallel channels of efficient thermal fillers elevate radial effective thermal conductivity to a new level.

3. **High-Conductivity Redundancy**
   Maintains conductivity close to 100% IACS (per International Annealed Copper Standard), reducing high-frequency loss at source.

PERFORMANCE TARGET

R_{th} Reduction > 40%

Significant coil temperature improvement vs. traditional

Thermal	Significantly lower radial thermal resistance
Electrical	Maintain high Q (Quality Factor)
Mechanical	Small-radius winding capable

03 Solution Advantages

Excellent Heat Dissipation

Radial thermal shortcut; breaks insulation thermal bottleneck.

Efficiency Gain

Lower ACR; maintain conversion efficiency at high frequency.

Integration & Scale

Compatible with existing coil fabrication and liquid cooling.

Long-Term Safety

Slows material thermal aging; extends charging terminal life.

Discuss Your Application

This solution is in prototype stage; Tier 1 suppliers welcome for technical alignment.

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